

A Convex-Order Characterization of All Dual Probabilistic Frames

A literature-implied full answer to the question in arXiv:2505.13885

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Status. Full characterization, obtained by combining conditional barycenters with Strassen’s martingale theorem. A measure is dual to a fixed probabilistic frame precisely when it is a convex-order mean-preserving spread of an admissible pushforward-type dual.

1 The source question

Chen, King, and Shonkwiler’s *Approximately Dual and Pseudo-Dual Probabilistic Frames* [1] defines a dual of a probabilistic frame $\mu \in \mathcal{P}_2(\mathbb{R}^n)$ to be a measure $\nu \in \mathcal{P}_2(\mathbb{R}^n)$ for which some coupling $\gamma \in \Gamma(\mu, \nu)$ satisfies

$$\int_{\mathbb{R}^n \times \mathbb{R}^n} xy^t d\gamma(x, y) = \text{Id}. \quad (1)$$

The Introduction (printed and PDF page 3) states that characterizing all such ν for a given μ remains open. The statement occurs in the paragraph beginning “Characterizing all the dual probabilistic frames”; the same challenge is repeated at the beginning of the section on approximately dual frames.

2 Exact characterization

Let

$$S_\mu := \int_{\mathbb{R}^n} xx^t d\mu(x)$$

be the positive definite frame operator, and define

$$\mathcal{H}_\mu := \left\{ h \in L^2(\mu; \mathbb{R}^n) : \int_{\mathbb{R}^n} xh(x)^t d\mu(x) = 0 \right\}.$$

For probability measures η, ν with finite first moments, write $\eta \preceq_{\text{cx}} \nu$ when $\int \varphi d\eta \leq \int \varphi d\nu$ for every convex φ for which the integrals are defined.

Theorem 1. *For every probabilistic frame μ on $\mathbb{R}^n$, its complete class of dual probabilistic frames is*

$$\mathcal{D}(\mu) = \bigcup_{h \in \mathcal{H}_\mu} \left\{ \nu \in \mathcal{P}_2(\mathbb{R}^n) : (S_\mu^{-1} \text{Id} + h)_\# \mu \preceq_{\text{cx}} \nu \right\}. \quad (2)$$

More explicitly, ν is dual to μ if and only if there are $h \in \mathcal{H}_\mu$ and a probability kernel $K_z(dy)$ such that, with $T(x) = S_\mu^{-1}x + h(x)$,

$$\int y K_z(dy) = z \quad \text{for } T_\# \mu\text{-a.e. } z, \quad \nu(B) = \int K_{T(x)}(B) d\mu(x).$$

Thus every dual is a mean-preserving spread of a pushforward-type dual, and every such spread is a dual.

Proof. Suppose first that ν is dual to μ , witnessed by γ . Disintegrate

$$d\gamma(x, y) = d\mu(x) K_x(dy)$$

and let $T(x) := \int y K_x(dy)$. Conditional Jensen and $\nu \in \mathcal{P}_2(\mathbb{R}^n)$ give $T \in L^2(\mu; \mathbb{R}^n)$ and

$$T_{\#}\mu \preceq_{\text{cx}} \nu.$$

Moreover, (1) becomes

$$\int x T(x)^t d\mu(x) = \text{Id}.$$

Put $h(x) = T(x) - S_{\mu}^{-1}x$. Since S_{μ} is symmetric,

$$\int x h(x)^t d\mu(x) = \text{Id} - S_{\mu}(S_{\mu}^{-1})^t = 0.$$

Hence $h \in \mathcal{H}_{\mu}$, proving the inclusion from left to right in (2).

Conversely, take $h \in \mathcal{H}_{\mu}$, put $T(x) = S_{\mu}^{-1}x + h(x)$ and $\eta = T_{\#}\mu$, and suppose $\eta \preceq_{\text{cx}} \nu$. Strassen's martingale theorem, in the explicit multidimensional kernel form [4, Theorem 2.3], supplies a probability kernel $K_z(dy)$ with first marginal η , second marginal ν , and barycenter $\int y K_z(dy) = z$. Define

$$d\gamma(x, y) := d\mu(x) K_{T(x)}(dy).$$

Then $\gamma \in \Gamma(\mu, \nu)$ and

$$\begin{aligned} \int xy^t d\gamma(x, y) &= \int x \left(\int y K_{T(x)}(dy) \right)^t d\mu(x) \\ &= \int x T(x)^t d\mu(x) \\ &= S_{\mu}(S_{\mu}^{-1})^t + \int x h(x)^t d\mu(x) = \text{Id}. \end{aligned}$$

Thus ν is a dual. (The same Cauchy–Schwarz argument used in the source then shows automatically that ν is a probabilistic frame.) $\square$

The point-mass case checks the orientation. If $\mu = \delta_a$ on $\mathbb{R}$ with $a \neq 0$, then $\mathcal{H}_{\mu} = \{0\}$ and the theorem says ν is dual exactly when $\delta_{1/a} \preceq_{\text{cx}} \nu$, equivalently when ν has mean $1/a$. This recovers the complete point-mass description in Chen and Schmoll [2, Proposition 5.12].

3 Provenance, search, and scope

This is a literature-implied answer. Strassen's 1965 theorem says that convex order is equivalent to the existence of a martingale coupling; it does not mention frames. The theorem is reproduced in explicit multidimensional kernel form as Theorem 2.3 in [4]. The source paper gives the pushforward condition (its Lemma 3.7), but does not pass from arbitrary couplings to conditional barycenters and convex order. The combination above is the direct identification that closes the stated characterization problem.

A bounded search through 13 August 2026 checked the full texts of arXiv:2505.13885 and arXiv:2501.02602 for the terms “convex order”, “martingale”, “Strassen”, “barycenter”, and “conditional expectation”, and searched the exact phrases “dual probabilistic frame convex order”, “transport dual

martingale”, and close variants. No prior statement of Theorem 1 was found. The source authors do not cite Strassen here and did not explicitly know they were posing an instance of his theorem. No priority claim is made.

The theorem completely characterizes which second marginal measures are duals, in every finite dimension. It does not classify extreme points of this nonclosed class, choose a canonical admissible h for a prescribed ν , or provide a numerical martingale-transport algorithm.

References

- [1] D. Chen, E. J. King, and C. Shonkwiler, *Approximately Dual and Pseudo-Dual Probabilistic Frames*, arXiv:2505.13885 (2025). See the Introduction, p. 3, and Lemma 3.7, p. 9.
- [2] D. Chen and M. Schmoll, *Probabilistic Frames and Wasserstein Distances*, arXiv:2501.02602v1 (2025). See Section 5 and Proposition 5.12.
- [3] V. Strassen, *The Existence of Probability Measures with Given Marginals*, Ann. Math. Statist. **36** (1965), 423–439. See Theorem 8.
- [4] K. Bołbotowski and G. Bouchitté, *Kantorovich–Rubinstein Duality Theory for the Hessian*, arXiv:2412.00516 (2024). See Theorem 2.3, printed and PDF page 15.